

Zhang Hui, Zhao Rui, Wang Zegao
 中国电子科技大学
 2007 Meritorious Winner

数学中国提供
<http://www.madio.net>

Team Control Number

1475

Problem Chosen

C

For office use only

T1 _____

T2 _____

T3 _____

T4 _____

For office use only

F1 _____

F2 _____

F3 _____

F4 _____

Summary

This paper is aimed to improve the current kidney transplant system in US and devise a procedure to maximize the number and quality of kidney exchange.

The length of the transplant candidate list reflects the effectiveness of the transplant system. We develop a model using linear regression to estimate the length of the candidate list. Based on the model, we predict the list will be 50000 people long in 2027. The model also indicates the potential bottlenecks for efficient kidney matching lies in the high price and side effect of immunosuppressive medications.

We find that scarcity of donor kidneys and low efficiency of donor-matching process are main reasons for the long candidate list. To increase the number of donor kidneys, we suggest US congress change the current “informed consent” policy to “presumed consent” policy, which has been successfully used in Spain. Using the previous model, we predict that the candidate list will begin to reduce in 2019 if the new policy can be used. If more resources were available, we suggest reduce the price of immunosuppressive medications to improve the efficiency of donor-matching efficiency.

We consider quality more important than number in kidney exchange, and establish a threshold value for kidney matching. Then, maximizing the number of exchanges turns to a “graph coloring problem”. It is estimated that about 1200 more annual transplants our procedure will generate.

In the paper submitted to HRSA, we argue against selling kidneys for transplantation and list our criteria used for priority and placement in allocating kidneys for the candidates.

By applying Analytic Hierarchy Process (AHP), we develop a model to estimate what factors are most important in influencing the potential donors’ decision. Based on the model, we find the probability of survival and future health risks for the donor are the main problems troubling the potential donors. So we suggest the government improve the technology of transplant operation and give the donors priority when they are on the transplant waiting list.

2007_ Mathematical Contest in Modeling (MCM) Control Sheet

Please review this page before submitting your solution to ensure that all of the information is correct Do not make changes by hand to the information on this control sheet. If you need to change any of the information on this sheet, login via the **Advisor Login** link on the MCM web site at

<http://www.comap.com/undergraduate/contests/mcm>, make the changes online, and print a new sheet.

You may NOT photocopy this control sheet to give to a new team, nor may you assign any team a control number. Each team must have its own control number, obtained by registering via the MCM web site at

<http://www.comap.com/undergraduate/contests/mcm>.

Advisor
Name: Li Mingqi

Department: Department of Biotechnology

Institution: School of Life Science and Technology in Univ. of Elec. Sci.
and Tech. of China

Address: No. 4, 2nd Section, Northern Jian She Road
Chengdu, Sichuan 610054

Phone: 86-28-83202166

Fax:

Email: mqli@uestc.edu.cn

Home Phone: 86-28-66098304

**Your team's control number
is:**

1475

(Place this control number on all
pages of your solution paper
and on any support material.)

Problem Chosen:

C

The names of the team members will appear on your team's certificate **exactly** as they appear on this page, including all capitalization and punctuation, if any. Gender data is optional and will be used for statistical purposes only; it will not appear on the certificate.

Team Member

Gender

Zhang Hui

M

Zhao Rui

M

Wang Zegao

M

Each team member must sign the statement below:

(Failure to obtain signatures from each team member will result in disqualification of the entire team.)

Each of us hereby testifies that our team abided by all of the contest's rules and did not consult with anyone who was not on this team in developing the enclosed solution paper.

Mailing Address and Signature of Zhang Hui

Mailing Address and Signature of Zhao Rui

Mailing Address and Signature of Wang Zegao

This signed original must be stapled to the top of one copy of your team's solution paper.

More Kidney Donors, More Lives Can Be Saved

Introduction

Nowadays, the demand for organs for transplantation drastically exceeds the number of donors. To help this situation, the Organ Procurement and Transplantation Network (OPTN) was established in 1984 to match organ donors to patients with organ needs. Today, the network is still functioning and more and more donors are identified and used each year. But the current system is headed for collapse with consequential failures for some of the neediest patients. Therefore, the network still remains to be improved.

Kidney (shown in Figure 1) plays an important role in human health. People with end-stage kidney disease face death, dialysis, or the hope for a kidney transplant. But many issues are involved in kidney transplantation. What's more, the transplanted kidney must be of the compatible ABO blood type and HLA. The donor and recipients mismatching makes a significant loss to the donor population. A kidney exchange system is useful to release the situation. So it's urgent to devise a procedure to maximize the number and quality of kidney exchanges. Meanwhile, when a kidney becomes available (may be barely compatible) from the cadaver queue, how do the patient decide whether to take an offered kidney, or to even participate in a kidney exchange?

Transplantation is a controversial issue with both technical and political issues. For instance, should transplants be performed only in a few large centers, by a few expert and experienced surgeons? Would that be a fair system and would it add or detract from system efficiency? Whether people selling organs for transplantation should be allowed? How to develop and recruit more altruistic donors? The problems listed above may require some changes to the criteria and policies.

Figure 1.The main components of kidney

Symbols

Table1.

Symbols used in this paper

Symbols	Definitions	Unite
$L(t)$	The length of the kidney transplant candidate list at the end of year t	people
$C(t)$	The number of new kidney transplant candidates placed on the candidate list in year t	people
$B(t)$	The number of kidney transplanted candidates in year t	people
N_d	The number of deceased kidney donors in year t	people
N_l	The number of living kidney donors in year t	people
$K(t)$	The number of kidneys can be used for transplantation in year t	
$P(t)$	The population of US in year t	people
$\gamma(t)$	Kidney donation rate in US in year t	PMP

General Assumptions

- Each living donor donates only one kidney and each deceased donor donates two kidneys.
- All the kidneys obtained from the donors (both living and deceased) can be used for kidney transplantation. Therefore, $K(t) = 2N_d(t) + N_l(t)$
- For the candidates who are eligible for transplantation, the transplant center offered only one kidney for each candidate.
- The population of US grows at a logistic form (ref. [1]).

Preparative

Definitions

- **Donation rate:** donors per million population (PMP), (ref. [9]).
- **New candidates:** the new patients who are placed on the candidate list.
- **Transplanted patients:** the patients who has received the transplantation operation.
- **Transplanted kidney:** the kidney which has been transplanted in the patient.

Matching Rules for Kidney Transplantation

It is helpful to divide donor and recipient matching into three distinct areas: blood type matching, tissue type matching and cross matching. For different organs, the criteria of organ matching are different. Here, we only consider the kidney matching rules in kidney transplantation.

- **Blood type matching.** There are four major blood types in humans, which are simply noted as blood type A, B, AB and O. In most circumstances, kidneys must be transplanted only into the candidates with the same blood type except in the case of zero antigen candidates with other blood type. However, the probability that the case of zero antigen candidates occur is very small (ref. [2]). Therefore, for simplicity, we assume the kidneys can only be transplanted into the candidates with the same blood type, shown in Figure 2.

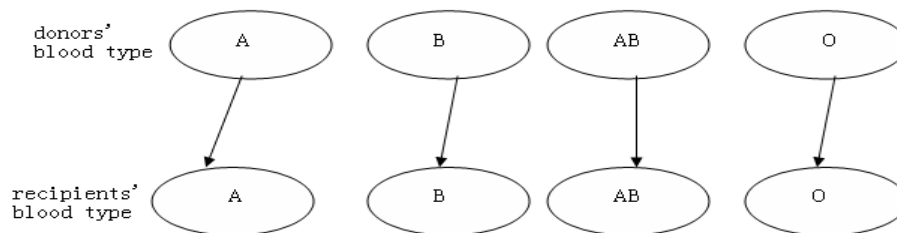


Figure 2.Blood type matching

- **Tissue matching.** Tissue matching is a very complex area involving testing the similarity of antigens between the donor and recipient. For kidney transplantation, we currently look at six of the antigens, called HLA antigens. By analyzing the HLA antigens both individuals have, we are able to determine the closeness of tissue matching. The requirement of tissue matching differs in different transplant center, usually at least 4 of 6 match (ref. [4]).

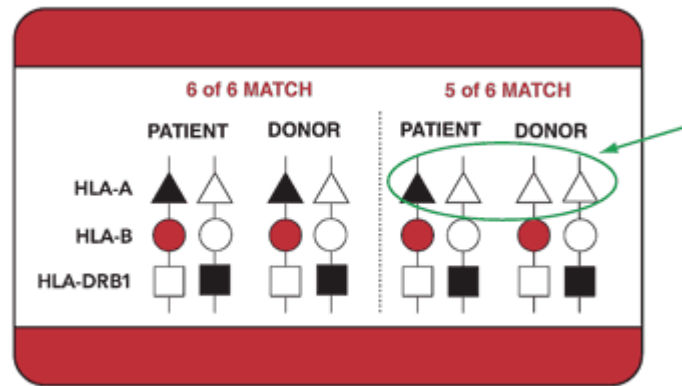


Figure 3. Tissue matching, 6 of 6 match and 5 of 6 match

- **Cross matching.** A positive cross match means that the recipient has responded to the donor and that the transplant should not be carried out. A negative cross match means that the recipient has not responded to the donor and therefore transplantation should be safe (ref. [4]).

Task 1: Modeling the Length of the Candidate List

We think the length of the transplant candidate list can well characterize the effectiveness of the transplant system, the longer the list is, the less effective the system is. This is reasonable, because the longer the candidate list is, the more time the candidates will spend in waiting for an available kidney for transplantation. But the waiting time is crucial for the candidates, because many patients can't live long enough to get a kidney for transplantation. It is reported that about 4000 candidates died during the waiting process (ref. [5]).

Length of the Candidate List

Consider the transplant candidate list to be a system with 2 inputs and 2 outputs (Figure 4). The two inputs are the new candidates placed on the list and the donors, the two outputs are the transplanted candidates and the candidates died because of lacking of available kidneys. Then the length of the candidate list can be given by

$$L(t) = L(t_0) + \sum_{t=t_0+1}^t C(t) - \sum_{t=t_0+1}^t B(t) - S \cdot (t - t_0) \quad (1)$$

Where

$L(t)$ = the length of the kidney transplant candidate list at the end of year t ,

$C(t)$ = the new candidates placed on the list in year t ,

$B(t)$ = the number of transplanted patients in year t ,

S = the number of candidates died per year, $S=4100$ people/year ([ref. [7]).

Equation (1) is the basic equation to characterize the change of the length of the candidate list. We can calculate the length of the candidate list once the functions $C(t)$ and $B(t)$ are determined.

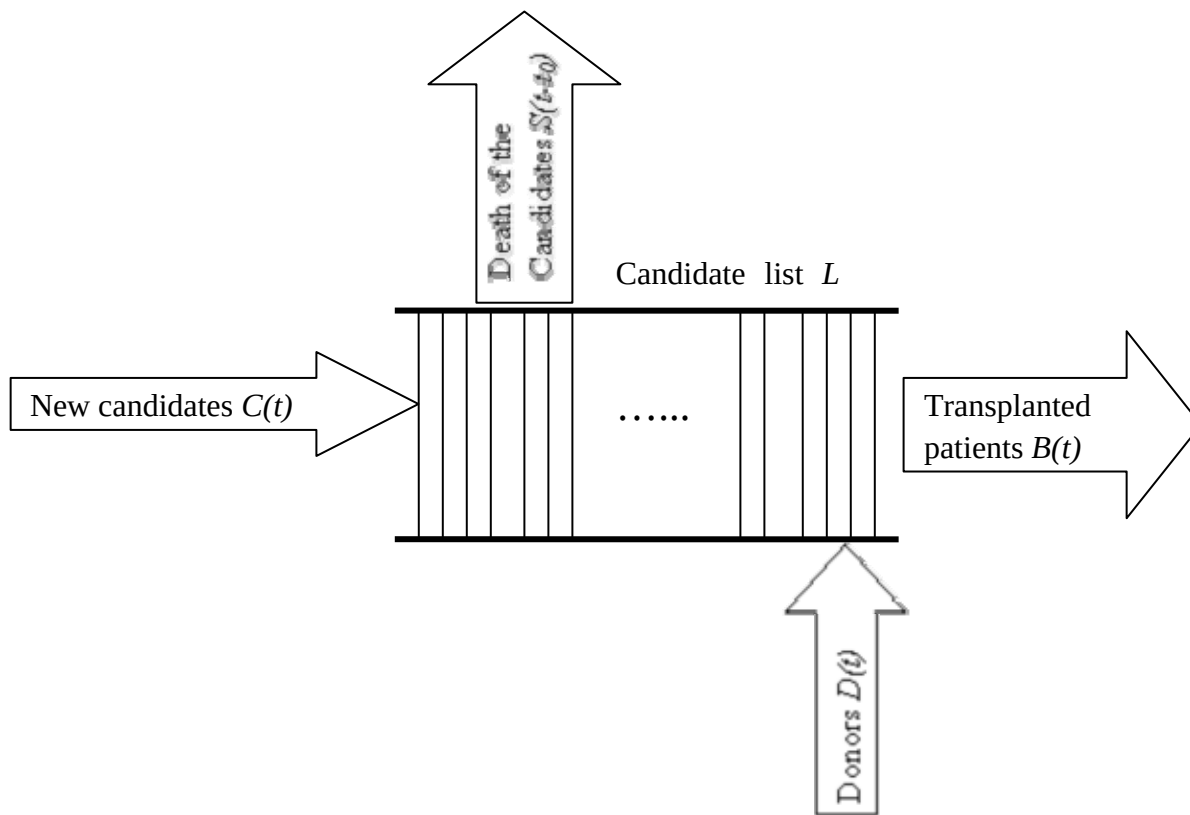


Figure 4. A system used to characterize the change of the length of the candidate list

Efficiency of Donor-matching Process

The relation of the number of transplanted patients and the number of kidney donors reflect the efficiency of donor-matching process. Some historical data about the kidney transplanted patients and the number of donor kidneys are shown in Table2.

Table2.

The number of donor kidneys and transplanted patients, 1999~2006, people

Year	1999	2000	2001	2002	2003	2004	2005	2006
Transplanted patients	12760	13612	14265	14779	15137	16004	16481	15721
Donor kidneys	15496	16471	17094	17516	17979	19297	19970	19131

The scatterplot graph indicate there may be a linear relation between the two variables, namely

$$B(t) = aK(t) + b$$

where

$B(t)$ = the number of transplanted patients in year t ,

$K(t)$ = the number of donor kidneys in year t .

We resort to linear regression to estimate the parameters in the above equation, yielding

$$B(t) = 0.83K(t) \quad (2)$$

with the multi correlation coefficient $R^2 = 0.98$. The high correlation indicates the regression equation

fits the data in Table 2 quite well, shown in Figure 5.

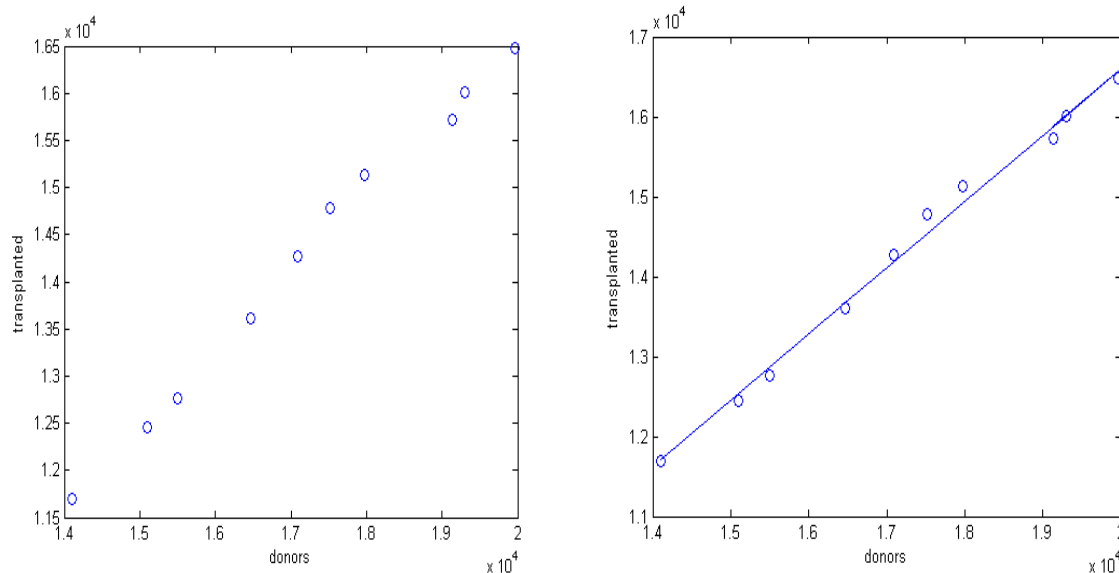


Figure 5. The scatterplot graph(left) and fitting the data in Table2 using linear regression(right).

Equation (2) means that the number of transplanted patients is in direct proportion to the number of donor kidneys, namely

$$B(t) = \beta \cdot K(t) \quad (3)$$

where β is a constant, which can be explained as the efficiency of donor-matching process, which is mainly determined by the organ matching probability. The value of organ matching probability is determined by the 3 matching rules listed in advance. For simplicity, we only consider the blood matching and tissue matching, then β is given by

$$\beta = P_B \cdot P_T$$

where

P_B = the probability that a donor can find a candidate on the list who is blood matching with him.

P_T = the probability that a donor can find a candidate on the list who is tissue matching with him.

In fact, the American blood type follows some certain distribution (Ref. [5]), so P_B must be a constant.

What's more, the type of the antigen HLA A, B, DR is certain (Ref. [6]), so P_T is also a constant.

Estimating the Number of New Candidates and Donor Kidneys

➤ The number of donor kidneys

To predict the number of donor kidneys, we collect some historical data, shown in Table 3.

Table 3.

The number of donors from 1999 to 2006, people

Year t	1999	2000	2001	2002	2003	2004	2005	2006
onor kidneys	15496	16471	17094	17516	17979	19297	19970	19131

Let $K(t)$ denote the number of donor kidneys in year t . We resort to linear regression to estimate the relation of $K(t)$ and t , yielding

$$K(t) = 626.28t - 1236281.6 \quad (4)$$

with the multi correlation coefficient $R^2 = 0.95$. The high correlation indicates the regression equation (4) fits the data in Table 3 quite well, depicted in Figure 6.

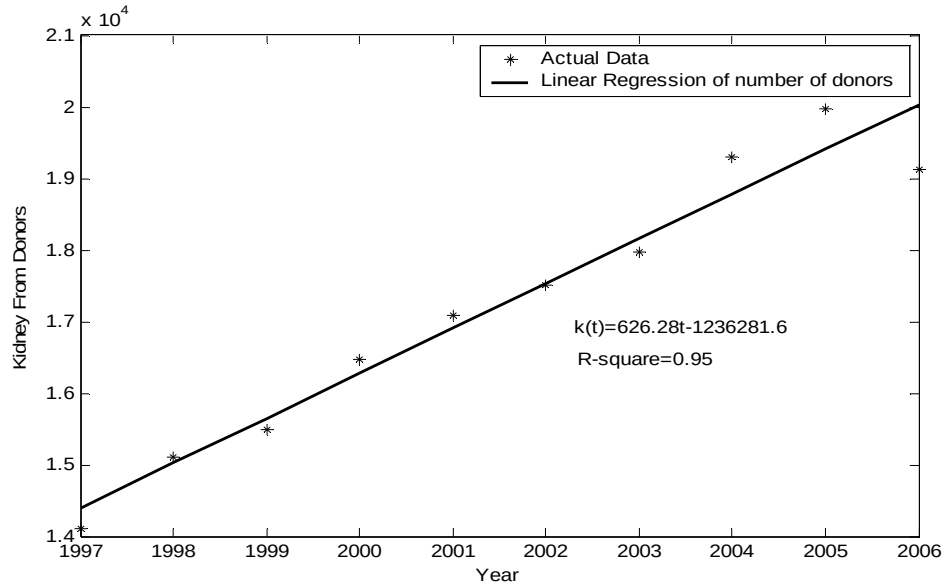


Figure 6. Fitting the data in Table 2 using linear regression (right)

➤ The Number of New Candidates

Some historical data about the number of new candidates from 1997 to 2006 are shown in Table 4.

Table 4.

The number of new candidates from 1999 to 2006, people

year	1999	2000	2001	2002	2003	2004	2005	2006
$C(t)$	22784	23420	23434	24613	25584	28308	30335	31239

Let $C(t)$ denote the number of new candidates in year t . We resort to linear regression to determine the relation of $C(t)$ and t , giving

$$C(t) = 1310.2t - 2597508.5 \quad (5)$$

with the multi correlation coefficient $R^2 = 0.96$. The high correlation indicates the regression equation (5) fits the data in Table 4 quite well, depicted in Figure 7.

Running the Model

We will run the model to predict the length of the candidate list.

Substituting equations (3), (4), (5) into equation (1) leads to

$$L(t) = L(t_0) + \sum_{t=t_0+1}^t (1310.2t - 2597508.5) - \beta \cdot \sum_{t=t_0+1}^t (626.28t - 1236281.6) - 4100(t - t_0)$$

(6)

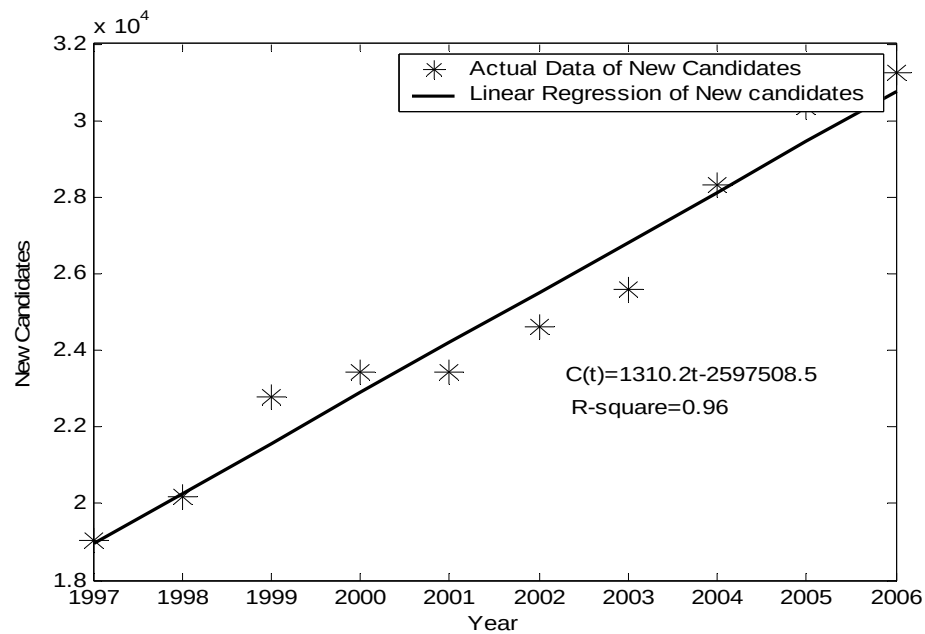


Figure 7. Fitting the data in Table 3 using linear regression (right)

What's more, it is reported that there are 69708 kidney transplant candidates at the end of 2006, namely $L(2006)=69708$. Set t_0 be 2006, equation (6) turn to

$$L(t) = 69708 + \sum_{t=2007}^t (1310.2t - 2597508.5) - \beta \cdot \sum_{t=2007}^t (626.28t - 1236281.6) - 4100(t - 2006) \quad (7)$$

In equation (7), the parameter β denotes the efficiency of donor-matching process. For different value of β , the change of the length of the candidate list is depicted in Figure 8.

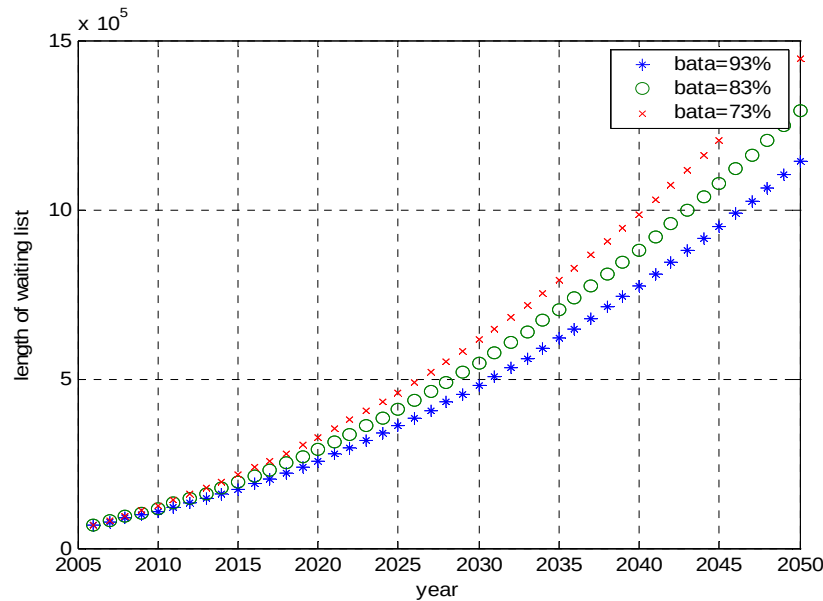


Figure 8. The length of the candidate list, for $\beta = 0.73, 0.83, 0.93$

The Figure showed that the higher the efficiency is, the slower the candidate list grows. Therefore, improving the efficiency of donor-matching is useful to shorten the candidate list.

Conclusions

➤ The current network is facing collapse because of the long candidate list.

The model indicates that the length of the candidate list is growing at a horrible rate. It will be 500000 people long in 2027. But it's believed that for any server of network, its capacity is limited. Therefore, the current server of the transplant network will collapse sooner or later no matter how big the capacity is, because the length of the list will increase all the time. To alleviate the network's high burden, **we suggest divide the current transplant work into some small networks.**

➤ Potential bottlenecks for efficient kidney matching.

The expression of kidney matching efficiency is given by $\beta = P_B \cdot P_T$. The current efficiency is only 0.83. To improve the efficiency of kidney matching, we should change the value of the 2 parameters in the formula.

The parameter P_B denotes the probability that a donor can find a candidate on the list who is blood matching with him. This parameter is hard to change, because the blood type follows some certain distribution (Ref. [5]), so the probability that the donors and the recipients are blood type matching must be a constant.

The parameter P_T denotes the probability that a donor can find a candidate on the list who is tissue matching with him. Usually, at least 4 of the 6 HLA antigens are required to be compatible between the donor and the recipient. However, the type of HLA antigens is about 1031(Ref. [6]), so it's quite hard to find a donor who are tissue matching with the recipient. However, the immunosuppressive medications used to prevent rejection have improved to the extent that even transplants with no tissue match (a zero match) may still provide good outcomes. But the immunosuppressive medications are quite expensive thereby many patients can't afford it. Besides, the medications have a bad effect in human health. Therefore, many patients opt to wait for a better matching donor, which waste quite a lot of donors.

In summary, the largest potential bottlenecks for efficient kidney matching are the high price and

side effect of the immunosuppressive medications.

Task 2: Changing the Current US Transplant Policy

Some Policies Used in Spain

Spain is widely perceived in the transplant community as having one of the best organ donation programs in the world, and many analysts believe that its **presumed consent policy** accounts for part of its success (Ref. [8]).

Presumed consent policy means that the government and procurement organizations may assume that citizens are willing to donate their organs at death if they do not state otherwise. There are two main kinds of presumed consent laws: “Strong” presumed consent laws rely solely on the individual citizen to declare himself or herself a non-donor. “Weak” presumed consent laws allow the family members of a decedent to opt-out if they desire to do so. Spain has a weak presumed consent law, where hospital staff members always approach the surviving family members—not to request permission to procure organs, but to see if they would prefer not to allow procurement to proceed as it normally would.

Spain also attributes some of its success to the **Active detection** policy. Active detection in this case means having transplant coordinators visit emergency rooms and the ICU on a daily basis, checking the roster of patients and their status. It is reported that Spain has seen its candidate list and waiting time reduced and its rate of donations increase. While its refusal rate from families is significantly lower than that in the U.S. The United Network Organs Sharing (UNOS) sees 21 donations PMP (per million population) at the end of 2006, while Spain saw donations as high as 39 PMP.

Model Refinement

➤ Estimating the population of US

It is assumed that the population of US grows in the logistic form (Ref. [1]), so we resort to logistic model to predict the population of US, namely

$$P(t) = \frac{P_M}{1 + \left(\frac{P_M}{P(t_0)} - 1 \right) \cdot \exp(-r \cdot (t - t_0))}$$

where

- t is time, with initial time $t_0 = 1950$,
- $P(t)$ is the population of US, in millions of people, with $P(t_0) = P(1950) = 157.8$,
- P_M is the environment capacity— in millions of people, we take $P_M = 500$,
- r is the intrinsic growth rate of the population, determined from data.

We use population data from 1950 to 2005 to fit the equation and get

$$P(t) = \frac{500}{1 + \left(\frac{500}{157.8} - 1 \right) \cdot \exp(-0.02 \cdot (t - t_0))} \quad (8)$$

with multi correlation coefficient $R^2 = 0.994$

➤ Estimating the length of the candidate list

Firstly, we assume each donor donate 2 kidneys. This is reasonable, because **presumed consent policy** means that the government and procurement organizations may assume that citizens are willing to donate their organs at death if they do not state otherwise, so most donors donate their kidneys after death.

Let $\gamma(t)$ denote the kidney donation rate in US in year t . What's more, we suppose the kidney donation rate grows at 4.5 PMP/per year, namely

$$\gamma(t) = 4.5 \times (t - 2006) + 21$$

Then we get

$$K(t) = \gamma \cdot P(t) = \frac{(9 \times (t - 2006) + 42) \times 500}{1 + \left(\frac{500}{157.8} - 1 \right) \cdot \exp(-0.02 \cdot (t - t_0))} \quad (9)$$

It follows from equations (1), (4), (5), (9) that

$$L(t) = 69708 + \sum_{t=2007}^t (1310.2t - 2597508.5) - 4100(t - 2006) - \beta \cdot \sum_{t=2009}^t \left(\frac{2 \cdot \gamma \cdot P_M}{1 + \left(\frac{P_M}{P(t_0)} - 1 \right) \exp(-r \cdot (t - t_0))} \right) \quad (10)$$

Suppose the value of β is 0.83, the change of the length of the candidate list is depicted in Figure 9.

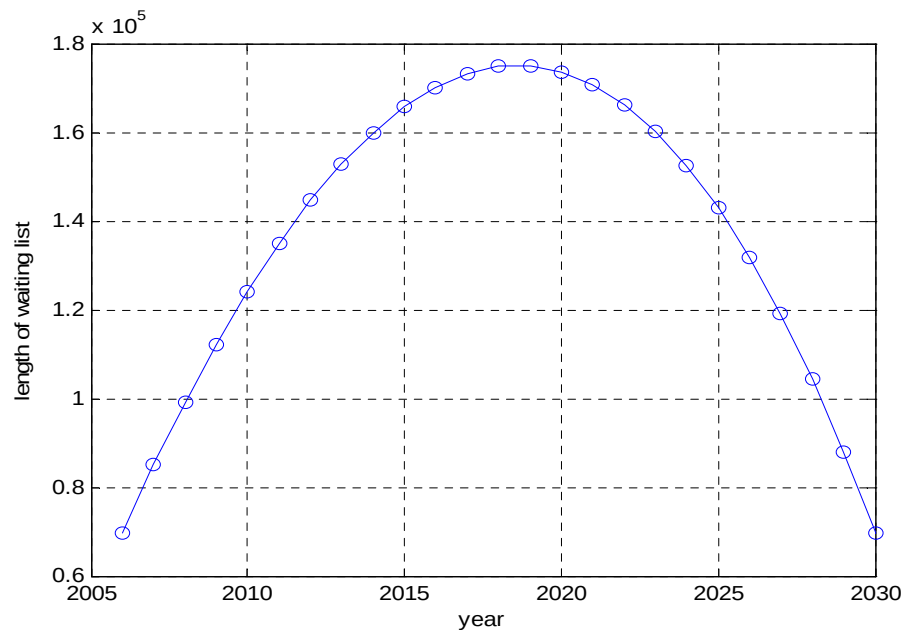


Figure 9. The length of the candidate list, for $\beta = 0.83$

Figure 9 shows that the length of the candidate list will increase at first, and begin to reduce in the year 2019. And the list will reduce to 0 about in year 2035. Therefore, the presumed consent policy is

very useful for US.

Report on Changing the Current Kidney Transplant Policies in US

To the US Congress:

Despite the continuing and dramatic advances in medicine and health technology, the demand for kidney transplantation drastically exceeds the number of donors. In order to know the current status of kidney transplantation in US and save more lives by kidney transplantation, we model the length of the kidney transplant candidate list. Our investigation indicates

- **The length of the candidate list is growing at a horrible rate.** By modeling the length of the candidate list, we find the length of the list is growing at a horrible rate. Every 12 minutes another name is added to the candidate list, and an average of 17 people die each day from the lack of compatible kidney for transplant.
- **The current network is facing collapse because of the long candidate list.** The model indicates that the length of the candidate list is growing at a horrible rate. It will be 500000 people long in 2027. But it's believed that for any server of network, its capacity is limited. Therefore, the current server of the transplant network will collapse sooner or later no matter how big the capacity is, because the length of the list will increase all the time. To alleviate the network's high burden, **we suggest divide the current transplant work into some small networks.**
- **Scarcity of donor kidneys and low efficiency of kidney matching** are two largest bottlenecks for kidney transplantation. Our model states that the high growth rate of the length of the candidate list is mainly caused by lacking of available kidneys for transplantation, which is determined by the number of donor kidneys and the efficiency of kidney matching.
- **The largest bottlenecks for efficient kidney matching lies in the high price and side effect of the immunosuppressive medications.** Our model indicates that the efficiency of kidney matching is mainly restricted by the low HLA matching probability. Although the immunosuppressive medications can improve the HLA matching probability theoretically, its high price and side effect reduce its efficiency

In order to obtain more donor kidneys and improve the efficiency of kidney matching, we suggest

- **Change the current “informed consent” policy to “presumed consent” policy.** We investigate the policies used in Spain, the country with the highest donation rate. Spain is widely perceived in the transplant community as having one of the best organ donation programs in the world, and many analysts believe that its **presumed consent policy** accounts for large part of its success. Presumed consent policy means that the government and procurement organizations may assume that citizens are willing to donate their organs at death if they do not state otherwise. By using presumed consent policy, the donation rate in Spain up to 39 PMP as of the end of 2006, much larger than 21 PMP in US (Ref. [8]).

Our model indicates that if the kidney donation rate in US can grow at a speed 4.5 PMP/year, the candidate list will begin to reduce in year 2019.

- **Allocate more resources for developing immunosuppressive medications.** It has been mentioned that the largest bottlenecks for efficient kidney matching lies in the high price and side effect of the immunosuppressive medications. If more resources are allocated for developing the immunosuppressive medications, the price of the medicine will come down so that more needy people can afford it, then the efficiency of kidney matching will raise.

Our model indicates that if the efficiency of kidney matching up to 0.93, with the donation rate grows at a speed 4.5PMP/year, the length of the candidate list will begin to reduce in year 2015.

- **Divide the current transplant network into some smaller networks.** The current transplant network is facing collapse because of the long candidate list and limited capacity of the server. So we suggest divide the current transplant work into some small networks to alleviate the burden of the main network.

Although the current status quo of kidney transplantation is not optimistic, we are full of confidence that the candidate list will be shorter and shorter if some current policies are to be changed.

Task 3: Modeling the Kidney Exchange

Definitions

- **Graph coloring:** Coloring a simple graph means using least number of colors to assign a color to each vertex of the graph so that no two adjacent vertices are assigned the same color.
- **Connected components:** A graph that is not connected is the union of two or more connected subgraphs, each pair of which has no vertex in common. These disjoint connected subgraphs are called the connected components.
- **DP-pair:** Pair of donor and patient.
- **Adjacent:** two vertices u and v in an undirected graph G are called adjacent in G if $\{u, v\}$ is an edge of G

Procedure to Maximize the Number of Exchanges

- **Step1: Building a directed graph**

Consider each DP-pair as a vertex in a directed graph. If the donor in DP-pair of vertex f is compatible with patient in DP-pair g , then there is a directed edge from vertex f to vertex g .

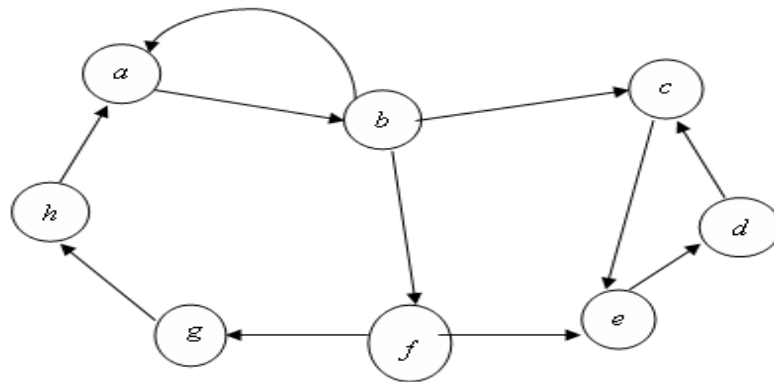


Figure 10. a directed graph, whose vertexes are DP-pairs

- **Step 2: Finding all of the circles in the directed graph**

Using Floyd's algorithm, we get all of the circles in the graph; possibly, some circles have common vertices. Take Figure 10 as an example, we have:

Circle A: $a \rightarrow b \rightarrow f \rightarrow g \rightarrow h \rightarrow a$.

Circle B: $a \rightarrow b \rightarrow a$.

Circle C: $c \rightarrow e \rightarrow d \rightarrow c$.

➤ Step 3:

A vertex is in one of these circles means the corresponding patient can find a compatible donor through exchange.

Maximizing the number of exchanges is equivalent to selecting some circles to maximize the total number of vertices, on condition that none of these vertices belong to two circles simultaneously.

Consider each circle as a set, the vertices in the circle as it's elements. The previous problem can be described as: select some sets to maximize the sum of cardinality of these sets, on condition that the intersection of any two sets is empty.

From the example above, we have:

Set $A = \{a, b, f, g, h\}$;

Set $B = \{a, b\}$;

Set $C = \{c, e, d\}$;

➤ Step 4: Building another undirected graph

Consider each set to be a vertex of an undirected graph, the cardinality of each set as the weight of the corresponding vertex. If $A \cap B \neq \emptyset$, then connect vertex A and vertex B . This means we can't select circle A and circle B simultaneously. The problem can be described as: select some vertices (each vertex stand for a set) from the undirected graph to maximize the total weight of these vertices, the vertices that adjacent can not be selected simultaneously.

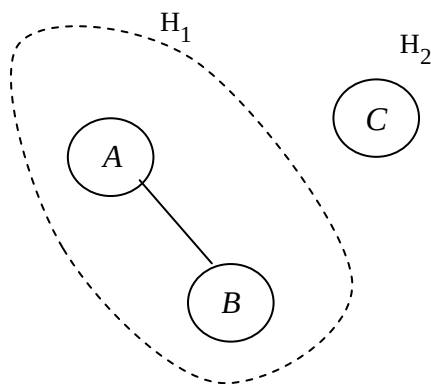


Figure11. The Graph H and Its Connected Components H_1, H_2 .

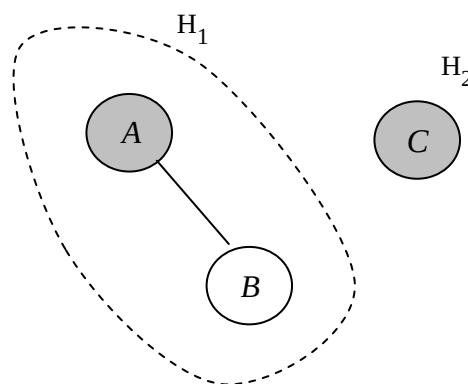


Figure 12. Coloring H_1 and H_2 separately.

Step 5: Coloring the undirected graph:

Considering every Connected Component separately, coloring it; sum the weight of vertices with the same color; find out the color that corresponds to the max number; the vertices with this color should be marked;

If vertex A was marked, the exchanges in circle A should be carried out;

Using step1 to step5 can maximize the number of exchanges.

In our example, vertex A and vertex C was marked, so the n-paired exchange in circle A and circle C should be carried out. We find 8 patients can find compatible donors through kidney exchange.

Conclusions

Using this exchange procedure and running the program in the Appendix, we find about 1200 transplants will be generated per year.

Task 4: Receive the Kidney or Not?

The transplant candidates usually face such an agonizing choice: when a barely compatible—in terms of HLA mismatches—kidney becomes available from the cadaver queue. Should they take it or wait for a better match kidney from the cadaver queue or from an exchange?

Model

The life of the transplanted kidney must be related with the compatibility between the donor and recipient, especially the terms of HLA matches (because blood type matching is essential for transplantation, but the terms of HLA matches can range from 0 mismatch to 6 mismatch). The level of HLA matching can be divided into 7 ranks: 0 mismatch, 1 mismatch, ..., 6 mismatch.

We can calculate the expected value of the transplanted kidney when the patient take the barely compatible kidney and participate in a kidney change, respectively. The strategy with larger expected value of the transplanted kidney will be the patient's choice.

Firstly, we define

- P_i be the probability that the recipient and the donor are i mismatch
- T_i be the life of the transplanted kidney which is obtained from the living donor who are i mismatch with the recipient.
- $T_i/2$ be the life of the transplanted kidney which is obtained from the deceased donor who are i mismatch with the recipient.
- $\bar{T}$ be the expected value of the kidney life if the patient take the barely compatible kidney.
- $\bar{T}'$ be the expected value of the kidney life if the patient participate in a kidney exchange.
- P_L be the probability that the patient can obtain a available kidney from a living donor if he/her participate in the kidney exchange.
- $1 - P_L$ be the probability that the patient can obtain a available kidney from a deceased donor if he/her participate in the kidney exchange.

Secondly, if the patient take the barely compatible kidney (5 mismatch or 6 mismatch), the expected value of the life of the transplanted kidney is given by

$$\bar{T}_i = P_i \cdot T_i, \quad i = 5 \text{ or } 6$$

Then, if the patient participate in a kidney exchange, the expected value of the life of the transplanted kidney is given by

$$\bar{T}' = P_L \cdot \sum_{i=0}^6 P_i \cdot T_i + (1 - P_L) \sum_{i=0}^6 P_i \cdot \frac{T_i}{2} \quad (11)$$

Equation (11) can be rewritten as

$$\bar{T}' = \frac{P_L + 1}{2} \cdot \sum_{i=0}^6 P_i \cdot T_i$$

Conclusions

To decide whether take the available kidney, we can compare the value of $\bar{T}$ and $\bar{T}'$

- If $\bar{T} \geq \bar{T}'$, take the kidney at once,
- If $\bar{T} < \bar{T}'$, participate in the kidney exchange will be better.

For the value of P_i , we can inquire the expert, such as the doctor on pathology, while T_i can be analyzed from the historical data and P_L is different because of different people's concept.

Task 5: A Paper to HRSA

Say No to Selling Organs for Transplantation

We resolutely oppose to sell human organs. The patients have the same right to life and the donors should have a better reason than profit.

Everyone is born equal, justice demands every person has an equal right to live. To protect this right, society has an obligation to ensure that every person, no matter rich or poor, has equal access to medical benefits. But if a market in organs were to develop, ability to pay would determine who could buy organs, while economic need would determine who would be motivated to sell their organs. The wealthy would end up organ buyers being sold by the poor. A market in organs would thus benefit the wealthy while putting pressures on the poor to endanger their own health. Such an unequal distribution of health benefits and burdens would be unjust.

Although many people would be more likely to donate if they could make a profit, the idea of selling human organs could cause many problems. For instance, placing a price on a human body part may cause more crime in which thieves are not trying to steal innocent people's material goods, but instead of their vital organs. Moreover, poor people may be willing to put themselves in mortal danger for the possibility of becoming rich. I believe the risks out weigh the rewards.

In summary, we must say no to selling organs for transplantation. What's more, we suggest that the following ethical principles and informed consent standards should be implemented for all living donors. The person who gives consent to becoming a live organ donor must be:

- competent (possessing decision making capacity)
- willing to donate
- free from coercion
- medically and psychosocially suitable
- fully informed of the risks and benefits as a donor
- fully informed of the risks, benefits, and alternative treatment available to the recipient.

To improve the organ transplant system to be more fair and efficient, the criteria used for priority and placement is important. We choose some criteria which we think is crucial in patient placement and rank order them,

- Waiting time; we place on the top of criteria because we think it's the most objective standard for patients ordering;
- Priority to local patient; it's convenient for patient and donor
- Priority to patient on good health condition; it will bring on higher overall survival and Less re-transplantation;
- Priority to Best biological match; it will bring on Higher overall survival; less re-transplantation;

Patient with a better biological match has a priority in current policies, but this may cause fewer transplants for certain groups, including highly sensitized patients and some minorities. We recommend the criteria and policy should concern these people.

Task 6: To Donate or Not To Donate? That's a Question

From the potential donor's perspective, the risks in volunteering involve assessing the probability of success for the recipient, the probability of survival of the donor, the probability of future health problems for the donor, the probability of future health risks (such as failure of the one remaining kidney), and the postoperative pain and recovery. How do these 5 factors influence the decision of the donor?

Model

Analytic Hierarchy Process (AHP for short) is quite effective to analyze this kind of problem. Notations used in the model are shown in Table 5.

Table 5.
Notations used in the model

Notations	Definitions
P	The probability that a potential donor reject to donate
P_1	The probability of success for the recipient
P_2	The probability of survival of the donor
P_3	The probability of future health problems for the donor
P_4	The probability of future health risks (such as failure of the one remaining kidney)
P_5	The postoperative pain and recovery

What's more, we let $F_1 \sim F_5$ denote the five factors listed in Table 5, respectively.

The weights of the five factors in influencing the potential donor's decision are determined by application of AHP. AHP is a measure to evaluate systems, which involves quantitative analysis and qualitative analysis. It exhibits the analytic and synthetically thoughts in decision-making strategy. AHP is quite suitable for the analysis of the affect decision model.

Firstly, the hierarchy of the problem is shown in Figure 13.

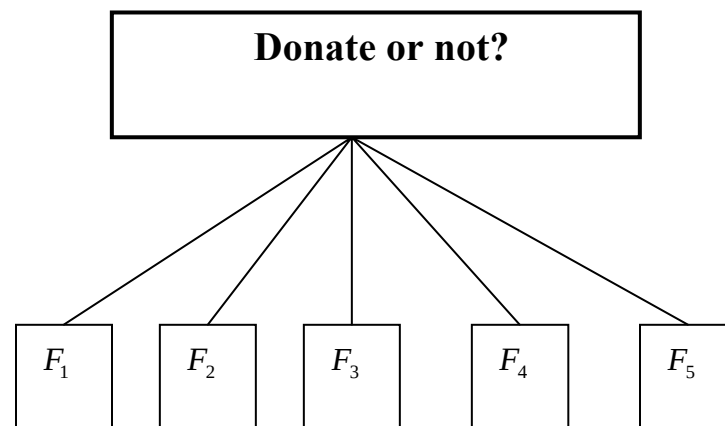


Figure 13.
Hierarchy of the factors

Our aim is to Figure out the weight of each factor in influencing the decision of the potential donor. We describe the calculating process as follows:

Basing on the analysis of the angst of the donor, we develop a judging matrix $\mathbf{P}$ to analyze the relative weight of each factor.

$$\mathbf{P} = \begin{pmatrix} 1 & P_{12} & P_{13} & P_{14} & P_{15} \\ P_{21} & 1 & P_{23} & P_{24} & P_{25} \\ P_{31} & P_{32} & 1 & P_{34} & P_{35} \\ P_{41} & P_{42} & P_{43} & 1 & P_{45} \\ P_{51} & P_{52} & P_{53} & P_{54} & 1 \end{pmatrix}$$

The element P_{mn} of the matrix represents the relative importance of factor F_m to factor F_n . Commonly we use 1, 2, 3...9 and their reciprocals to represent different degrees of importance. The bigger the number is, the more important it is. Since P_{mn} represents the importance of factor F_m to factor F_n , so the importance of factor F_n to factor F_m is $1/P_{mn}$.

Secondly, we normalize the column vectors in the judging matrix:

$$\overline{P_{mn}} = \frac{P_{mn}}{\sum_{k=1}^5 P_{kn}} \quad (m, n = 1, 2, 3, 4, 5)$$

Then add the normalized matrix in rows:

$$\overline{\alpha_m} = \sum_{n=1}^5 \overline{P_{mn}} \quad (m, n = 1, 2, 3, 4, 5)$$

The result should be normalized again to get the eigenvector:

$$\beta_m = \frac{\overline{\alpha_m}}{\sum_{k=1}^5 \overline{\alpha_k}} \quad (m = 1, 2, 3, 4, 5)$$

The eigenvector β represents the five factors' weights in influencing the potential donors' decision.

Use the judging matrix $\mathbf{P}$ and eigenvector β to calculate the maximum latent root:

$$\lambda_{\max} = \sum \frac{(\mathbf{P} \cdot \boldsymbol{\alpha})_m}{5\alpha_m}$$

where $(\mathbf{P} \cdot \boldsymbol{\alpha})_m$ represents the m th element of vector $(\mathbf{P} \cdot \boldsymbol{\alpha})$.

At last, we check the coherence of the judging matrix. The standard of coherence CI is given by:

$$CI = \frac{\lambda_{\max} - 5}{4}$$

For a 5-rank matrix, if $CI < 0.112$, the coherence of the judging matrix is suitable. Otherwise, the judging matrix needs to be adjusted (Ref. [10]).

Solving the Model

We determine the relative importance of the 5 factors in influencing the donors' decision experimentally and get the judging matrix

$$\mathbf{P} = \begin{Bmatrix} 1 & 1/7 & 1/3 & 1/4 & 1/2 \\ 7/1 & 1 & 5/1 & 3/2 & 6/1 \\ 3/1 & 1/5 & 1 & 1/4 & 2/3 \\ 4/1 & 2/3 & 4/1 & 1 & 3/1 \\ 2/1 & 1/6 & 3/2 & 1/3 & 1 \end{Bmatrix}$$

Following the approach, we Figure out the eigenvector of the factor judging matrix as:

$$\beta = (0.0948 \quad 0.8140 \quad 0.1685 \quad 0.5169 \quad 0.1813)$$

The standard of coherence CI is 0.0310 which satisfy the requirement $CI < 0.112$, so the judging matrix is effective.

What's more, for the eigenvector β , two conditions must be fulfilled:

- The elements in matrix R are positive numbers
- The sum of the elements in matrix R equals 1.

which is called "normalize the eigenvector", so maybe some adjustment is needed to make sure the conditions. At last, we get normalized eigenvector β as:

$$\beta = (0.0534 \quad 0.4585 \quad 0.0949 \quad 0.02911 \quad 0.1021)$$

Then, the probability of the donor contributing P :

$$P = 0.0534 \times P_1 + 0.4585 \times P_2 + 0.0949 \times P_3 + 0.02911 \times P_4 + 0.1021 \times P_5$$

Based on the expression, we conclude that the probability of survival for the donor is the most important factor influencing the decision of the potential donor, while the probability of future health risks (such as failure of the one remaining kidney) is the second, and so on.

Ways to Recruit More Altruistic Donors

Nowadays, the candidate list is becoming longer and longer, scarcity of donor kidneys is the largest bottleneck for kidney transplantation. It's estimated that about 17 patients in the candidate list died of lacking available kidneys each day. Therefore, it's very urgent to recruit more altruistic donors. Based on our model, we suggest

- **Developing the technology of transplantation operation.** Our model indicates that the probability of survival for the donor is the most important factor influencing the decision of the potential donor. Therefore, if the technology of transplantation operation developed, more people will be donors.
- **Propagandize the donation will not effect human health.** Our model indicates that the probability of future health risks plays an important role in the factors that influencing the potential donors' decision. In fact, it's reported that there will not be any effect for the health people to donate one of their kidneys. But many people don't know the fact, so propagandize to the donors the unharmed of the donation will be effective to increase the number of kidneys.
- **Allocate some compensation resources for the donors.** Our model indicates that the postoperative pain and recovery is also the potential donors cared about. Although we don't support the donors to ask compensation for their donation, we still think the government should allocate some resources for the donors in order to make the donors recover well after the operation.
- **Give priority to the donors when they are on the candidate list.** For the donors, if they are on the candidate list, give them priority in order to compensate for their donation. Many more people will be donors there after.

References

- [1] Alfred Clark, Jr. **2006. Logistic Model for US Population.** Retrieved February 12, 2007, from <http://www.me.rochester.edu/courses/ME406/webexamp/logisticde.pdf>
- [2] UCSF Medical Center. 2006. Kidney Transplant. Retrieved February 12, 2007, from http://www.ucsfhealth.org/adult/medical_services/organ_transplants/kidney/conditions/kidney/diagn
- [3]AAKP. 2006. Make Me a Perfect Match: Understanding Transplant Compatibility. Retrieved February 12, 2007, from <http://www.aakp.org/aakp-library/Transplant-Compatibility/index.cfm>
- [4]AAKP. 2006. Kidney Transplant Matching: What It Means. Retrieved February 12, 2007, from <http://www.aakp.org/aakp-library/Kidney-Transplant-Matching/>
- [5] Reference Weekly. 2005. The Blood Type Rate of World People. Retrieved February 12, 2007, from <http://naturegarden.bokee.com/3401368.html>
- [6]China Network. 2006. Three Generts be Discovered in Shenzhen. Retrieved February 12, 2007, from <http://www.china.com.cn/chinese/difang/244219.html>
- [7] The Transplant Network. 2006. Critical Data. Retrieved February 12, 2007, from <http://www.thetransplantnetwork.com/criticaldata.shtml>
- [8]Sam Crowe and Eric Cohen. 2006. Organ Transplant Polices and Policy Reforms. Retrieved February 12, 2007, from <http://www.bioethics.gov/background/crowepaper.htm>
- [9]Lewin. 1998. Performance Indicators. Retrieved February 12, 2007, from http://aspe.hhs.gov/health/orgdonor/evalrpt/performance_indicators.htm
- [10]Qiyuan Jiang. Mathematic Model. Beijing: Advanced Education Public. 2003. 224~243.

Appendix

The program for finding the maximum donor-recipient matches:

This program is written in C++.

```
var
  n,s,t,i,j:integer;
  dist:array[1..6000,1..6000] of real;
  prev:array[1..6000,1..6000] of integer;
procedure init;
var
  m,i,u,v:integer;
begin
  assign(input,'c:\maxmatching.txt');
  reset(input);
  assign(output,'c:\maxmatching.txt');
  rewrite(output);
  readln(n,m);
  fillchar(prev,sizeof(prev),0);
  for u:=1 to n do
    for v:=1 to n do
      dist[u,v]:=1e10;
  for i:=1 to m do
    begin
      readln(u,v);
      dist[u,v]:=1;
      prev[u,v]:=u;
    end;
  end;
procedure floyd;
var
  i,j,k:integer;
begin
  for k:=1 to n do
    for i:=1 to n do
```

```
    for j:=1 to n do
      if (dist[i,k]+dist[k,j]<dist[i,j])
        then begin
          dist[i,j]:=dist[i,k]+dist[k,j];
          prev[i,j]:=prev[k,j];
        end;
    end;{floyd}
procedure print(i,j:integer);
begin
  if i=j
    then write(i)
    else if prev[i,j]=0
      then write('No Circle!')
      else begin
        print(i,prev[i,j]);
        write(' ',j);
      end;
  end;{print}
begin
  init;
  floyd;
  for i:=1 to n do
    begin
      if dist[i,i]<>1e10
        then
          begin
            print(i,prev[i,i]);
            writeln;
          end;
    end;
  close(input);
  close(output);
end.
```